

A hybrid metaheuristic for smart waste collection problems with workload concerns

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ABSTRACT

In this paper, a hybrid metaheuristic is developed to solve the smart waste collection problem with workload concerns. It is composed by: (i) a look-ahead heuristic aiming at deciding the days in which collection is necessary and which bins need to be collected (must-go) considering the present bin fill levels and future bin fill level predictions; and (ii) a simulated annealing/neighborhood search algorithm to choose the bins that are profitable to collect and the best route(s) to visit the bins. This algorithm was developed to find solutions within a relatively short amount of time (we considered two hours as reference), as required for practical operations. The proposed hybrid metaheuristic is applied to randomly-generated test instances of sizes up to 500 bins and to a real case study of recyclables collection, leading to results that demonstrate its effectiveness and usefulness in practice when dealing with large-size instances. For the real case study, involving a major waste management company in Portugal, the profit achieved by the hybrid metaheuristic is at least 45% higher than the profit obtained by the company, and, at the same time, compliance with maximum shift duration and route workload balance is clearly better when the metaheuristic is used than in current operations.

1. Introduction

In recent years, transportation and logistics systems are being revolutionized by the Internet of Things (IoT) due to the rise of data availability on a near-continuous basis. The existence of these data offers new opportunities to enhance the performance of such systems, yet, at the same time, raises additional questions about how to design operations that will be profitable, environmentally friendly and socially fair. This is becoming a reality in waste collection management, which is the focus of this paper. Some technological companies have been developing sensors to place inside waste bins with the goal of detecting bin fill levels across the day. In addition to the bin fill levels, historical sensor data also allow obtaining bin accumulation rates with higher accuracy. Knowing these data constitutes an important step towards the development of decision-support approaches that not only “know” the present but are also able to “look” into the future. This makes possible the transition from “blind” to “smart” collection operations.

In blind collection operations, routes and route frequencies are pre-

defined according to the number of bins in the route and their estimated accumulation rates without knowing the real fill levels of the bins (Ramos et al., 2014a, 2014b; Bing et al., 2016). On the one hand, this implies that a potentially large percentage of visited bins presents a low fill level, causing unnecessary kilometers to be traveled and increasing waste collection costs as well as environmental pollution and greenhouse gas emissions. On the other hand, there is the possibility of bin overflows, which have a highly negative impact on public health and city image and are very much unwanted by municipal authorities. Indeed, one of the main concerns of waste collection companies is to avoid waste bins overflows. On the contrary, “smart collection” is dynamic and makes possible to determine the bins that need to be collected at each day considering their real fill levels and optimizing the routes accordingly, thus circumventing the disadvantages of “blind collection”.

Handling smart waste collection systems through exact methods is difficult, not only because of their sizes (involving several hundreds or even thousands of bins), but also because their solutions should be found within a relatively short amount of time (we took two hours as

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reference). This is necessary because data sent by the sensors upon which decisions are made should be as much up to date as possible, and the routes generated for the waste collection vehicles have to be checked and validated before being communicated to the collection teams in order to be realized later that day (during the night) or in the morning of the next day. Previous work as the one presented in Ramos et al. (2018a), where only bin and collection vehicle capacity constraints were considered, has shown that exact optimization algorithms may not be suitable to solve those large instances. Despite being able to provide much better solutions than the ones waste collection companies put into practice currently, they may not be fast enough considering the time-frame companies have to define the routes. For a case study with 226 bins, the exact branch-and-cut algorithm (GAMS-CPLEX) applied by Ramos et al. (2018a) ran during 4 h and could not find a guaranteed optimal solution. For larger instances and considering workload concerns, it may even be impossible to find a single feasible solution. Metaheuristic approaches, despite not ensuring optimal solutions as exact optimization algorithms do when their execution can be completed on time, may be instead a good option, as they may generate good-quality solutions in an acceptable amount of time, while meeting all operational constraints. In this context, our objective is to develop a hybrid metaheuristic approach for tackling real cases of smart waste collection systems given the real-time data provided by sensors.

In short, the main contributions of this paper are:

- The development of a hybrid metaheuristic to solve large-size smart waste collection problems consisting of:
 - o a look-ahead heuristic (that takes into consideration what is expected to happen in the future) to select the days when bin collections are necessary and the must-go bins in these days; and
 - o a simulated annealing/neighborhood search algorithm to decide which bins are worth to be collected and the best routes to visit the bins.
- The inclusion of workload concerns (maximum shift duration, and route workload balance) within smart waste collection problems, which is essential to accurately represent reality (such concerns have rarely been considered even in blind waste collection problems).
- The application of the hybrid metaheuristic to a real smart recyclable waste collection problem in a computational time compatible with the needs of practice and with results that significantly improve on those of actual operations.

Despite being applied to the smart waste collection problem in this paper, the proposed hybrid metaheuristic approach is generic in scope and can be easily adapted to handle any other smart collection problems (in particular, its parameters may have to be recalibrated).

The remainder of this paper is organized as follows. In Section 2, we present a description of the Smart Waste Collection Routing Problem with workload concerns. Section 3 consists of a review of related work: Vehicle Routing Problems with Profits (VRPPs), dynamic waste collection models, workload concerns, and hybrid metaheuristics in vehicle routing problems. Section 4 contains an extensive description of the proposed hybrid metaheuristic approach. In Section 5, we provide a detailed computational study of the application of this metaheuristic to a large set of randomly-generated instances. Section 6 presents the real case study data and the results obtained through our hybrid metaheuristic. Finally, Section 7 concludes the paper with a summary of our work and suggestions for future research.

2. Problem description

The problem addressed in this paper is an extension of the Smart Waste Collection Routing Problem considering the use of real-time information about bin fill levels and its expected evolution in the following days to define dynamic routes, and taking into account workload concerns. It is an operational problem faced by waste management

companies, whose tactical and strategic decisions involving the number and capacity of vehicles, the number, location and capacity of bins, and the depot location(s) have been taken and implemented beforehand.

This problem is defined based on a set of B waste bins, a set of V homogeneous vehicles and the depot(s) where all vehicles start and end their routes, a complete undirected graph connecting the bins and the depot(s), with distances d_{ij} for each edge (i,j) in the graph. Each waste bin i has a maximum capacity, E_i . A travelling cost c per unit of distance travelled as well as a revenue value r per kg of waste collected are considered. Each vehicle has a fixed capacity Q (in kg). At each day t , at a given time, the sensors located inside the waste bins transmit information on bin fill levels, in m^3 , and this information is translated into kg given the waste density (ρ). To simulate the data transmission process, two parameters characterize each bin i : a stock, which corresponds to the data read and transmitted by the sensors for each bin i at day t (W_{it}), and an expected daily accumulation rate, which is defined based on historical data read by the sensors for each bin i at day t (a_{it}).

For each day, the first decision to be taken is to determine whether collection operations must be performed in that day or not, which depends only on the existence of full (or near-full) bins (designated as must-go bins). The must-go bins must be collected to avoid overflows. If there are full or overflowing bins, other decisions need to be taken, namely: which other bins are worth being collected; and in which route (s) and sequence(s) should the must-go bins and these other bins be collected. The time to make these decisions is short (say, two hours), in order to consider bin fill data that are as much as possible up to date, and to account for the time needed to check and validate the routes before being communicated to the teams that perform the collection afterwards.

These decisions are made each day by the company with the aim of maximizing profit, while also maximizing route workload balance and complying with the maximum shift duration of collection teams, collection vehicle capacity and bin capacity. Profit is obtained by subtracting the costs associated to the collection operations from the revenues made by selling the waste collected (i.e., the waste has some economic value, as is the case of recyclables or waste that is subsequently used for energy production). As part of the bins to include in the routes are selected based on waste quantities that are known or predicted for a number of following days (the period of analysis), maximizing profit signifies anticipating profit as much as possible, which makes sense from the perspective of financial analysis. Workload balance is obtained by the difference between the duration of the longest route and the duration of the shortest route.

Workload concerns (maximum shift duration and route workload balance) have been dealt with in a very limited number of papers on waste collection. However, they are critically important to guarantee, on the one hand, that the company complies with labor legislation and avoids the often-unnecessary costs with employees' extra working time (when it is allowed by law), and, on the other hand, that work effort is evenly distributed across collection teams.

3. Related work

The Smart Waste Collection Routing Problem dealt with in this paper fits into the large body of literature concerning waste collection and vehicle routing problems (VRPs). Several recent surveys are available about both subjects taken together or separately. With respect to VRPs, we highlight the reviews (and taxonomies) proposed in Toth and Vigo (2014) and Konstantakopoulos et al. (2020). Waste collection management and/or routing problems have been surveyed by Beliën et al. (2014), Han and Ponce Cueto (2015) and Hannan et al. (2020). Within this body of literature, two other relevant reviews are due to Melaré et al. (2017) and Esmailian et al. (2018), respectively focused on technologies and decision support systems, and on the future of waste management in smart and sustainable cities. In both cases, the authors emphasize the role that sensor-based solutions can play in support of

collection operations.

The literature discussed in the publications quoted above liaises with our work to some extent. However, in this section, we will only focus on topics of specific interest for this paper. These topics are as follows:

- Vehicle routing problems with profits (VRPPs)
- Dynamic waste collection models
- Workload concerns (in VRPs)
- Hybrid metaheuristics in VRPs

3.1. Vehicle routing problems with profits

There are several variants of VRPPs according to the objective function, constraints and the number of vehicles considered. Bearing in mind the single-vehicle case, if the objective is just the maximization of revenues, the VRPP is called Orienteering Problem (also known as Selective Traveling Salesperson Problem, Maximum Collection Problem and Bank Robber Problem). The Orienteering Problem (OP) includes maximum shift duration constraints. If the objective function corresponds to the maximization of the difference between revenues and costs, the VRPP is called Profitable Tour Problem (PTP). In this case, no maximum shift duration constraints are included. There is also the Prize-Collecting Traveling Salesperson Problem in which the objective is the minimization of costs, but there is a constraint on the route revenues (i. e., they should be above a given threshold). Regarding the multiple-vehicle case, there are the Team Orienteering Problem, which is similar to the OP, but with more than one vehicle, and the Capacitated Profitable Tour Problem (CPTP), which is similar to the PTP, but includes vehicle capacity constraints. A literature review on the VRPP variants can be found in [Archetti et al. \(2014\)](#).

Most of the existing literature is dedicated to single-vehicle VRPP variants. However, our focus in this paper is placed on the CPTP, since our problem is characterized by multiple capacitated vehicles and our goal is the maximization of profit (difference between revenues and costs) as described in the previous section. Literature on the CPTP is scarce and recent. The first article addressing this problem is due to [Archetti et al. \(2009\)](#). These authors applied Tabu Search and Variable Neighborhood Search to solve CPTP instances with up to 199 nodes, concluding that the performance of both metaheuristics was similar. [Archetti et al. \(2013\)](#) presented a branch-and-price algorithm together with a restrictive master heuristic to solve the same problem, concluding, for the same benchmark instances, that this algorithm performed better than the metaheuristics of [Archetti et al. \(2009\)](#) in four of the instances (out of ten). Several extensions of the CPTP have been studied later. In 2016, [Li et al. \(2016a, 2016b\)](#) developed two approaches to solve the CPTP with time windows, namely a mixed-integer linear programming model (MILP) and an adaptive large neighborhood search (ALNS). The MILP model and the ALNS approach were evaluated on 54 randomly-generated instances with 10–100 nodes. Results show that the ALNS performs significantly better than the MILP model in terms of solution quality and CPU time. [Archetti et al. \(2017\)](#) developed a two-phase branch-and-cut algorithm to solve a CPTP with maximum shift duration constraints. This algorithm was applied to instances with up to 50 nodes, showing to be effective in optimally solving a large number of instances. [Stavropoulou et al. \(2019\)](#) proposed an Adaptive Tabu Search algorithm to solve a CPTP with maximum shift duration constraints and periodic deliveries (of the same quantities and by the same person). It was applied to benchmark instances as well as to newly generated instances with up to 200 nodes, achieving optimal solutions for small instances and performing well for larger instances. Very recently, [Orlís et al. \(2020\)](#) developed a branch-and-cut algorithm to solve a CPTP, applying it to randomly-generated instances with up to 30 nodes and to a real case study of a Dutch cash supply chain with a maximum of 35 nodes. Results indicate that the algorithm was able to solve a significant number of the generated instances within a two-hour

runtime limit and a total average gap of 9.6%. Regarding the case study, six out of the seven instances were solved close to optimality.

As it can be noticed, the literature on the problem described in this paper corresponding to a VRPP (CPTP) with workload concerns is quite meager. In addition, the existing one addresses only small size instances that are not realistic when compared to the real-world problems of waste collection that we want to address.

3.2. Dynamic waste collection models

Dynamic vehicle routing models have seldom been applied to waste collection problems. The first applications appeared only in the last decade, probably because the decision on which bins to collect is highly dependent on the existence of real-time information on bin fill levels and accumulation rates. Obtaining this information has only become viable in the last years with the emergence of data acquisition technologies, such as sensors, and its increasing availability.

Despite this, there are three papers where dynamic waste collection is studied based only on historical data (thus without considering real-time information): [Anghinolfi et al. \(2013\)](#), [Abdallah et al. \(2019\)](#), and [Exposito-Marquez et al. \(2019\)](#). [Anghinolfi et al. \(2013\)](#) solved a dynamic optimization model for waste collection route definition. Bins were grouped in macro cells inside which the paths between bins were fixed and known. Only three macro cells having fixed paths were considered, which resulted in a very rough simplification of reality. Recently, [Abdallah et al. \(2019\)](#) proposed an algorithm to select bins to collect based on historical fill rates. The selected bins are the input to the Dijkstra's algorithm that optimizes the routes. This methodology was applied to a realistic (inspired by reality) case study, corresponding to one of the largest neighborhoods in the city of Abu Dhabi (United Arab Emirates), with 658 bins divided into 25 sets. Also, in 2019, [Exposito-Marquez et al. \(2019\)](#) developed a greedy randomized adaptive search procedure to select bins and optimize routes for recyclable waste collection. The objective considered in this procedure was the minimization of the number of bins visited, i. e., only the bins with the highest fill levels should be visited. Moreover, they assumed unlimited capacity for vehicles. Maximum shift duration constraints were taken into consideration by these authors. The proposed procedure was applied to a case study in La Palma, Canary Islands, Spain, with 399 containers for plastic and 375 containers for paper/cardboard.

Relying on data acquisition technologies and real-time information, several studies on dynamic waste collection have been published in the last five years ([Anagnostopoulos et al., 2015](#); [Gutierrez et al., 2015](#); [Ramos et al., 2018a, 2018b](#); [Hrabeca et al., 2019](#)). [Anagnostopoulos et al. \(2015\)](#) studied a dynamic waste collection architecture based on data retrieved by sensors. In the system proposed, routes are defined for the regular (usually collected) bins. If during these routes, a bin becomes full or its fill level is over a pre-defined threshold, a routing algorithm is triggered to calculate a new optimal route including this critical bin, and drivers are forced to divert from the original route in order to collect this bin. In the same year, [Gutierrez et al. \(2015\)](#) proposed an artificial intelligence model, based on sensor data stored in a database, to select which bins to include in the routes. After this selection, routes are optimized in order to minimize the distance travelled. Both decisions are made independently of each other. [Ramos et al. \(2018b\)](#) proposed an optimization model to solve daily a real dynamic waste collection problem (selecting bins and defining routes) and applied it to a small case study with 68 bins. The same authors ([Ramos et al., 2018a](#)) extended the previous model to include two new parameters, the percentage of bins that can overflow, and the maximum allowed overflow threshold. In this extended model, a constraint on the maximum number of available collection vehicles was also added. As already mentioned, this model was applied to a real case study with 226 bins. More recently, [Hrabeca et al. \(2019\)](#) proposed a dynamic waste collection optimization model that, by making use of sensors data, selects the bins to be collected and the optimal routes for a collection period of one week. Two

objectives were considered in this model: the first one is to minimize transport costs and the second one is to optimally decide on early collection of partly-filled bins. This model was applied to a very small case study with only ten bins.

With the exception of [Expósito-Márquez et al. \(2019\)](#), that considers maximum shift duration constraints (but not the maximization of route workload balance) and tackles instances of significant size, none of the previous papers deals with large instances or considers workload concerns.

3.3. Workload concerns in VRPs

Workload concerns are important to make model formulations applicable to real-world problems, and particularly waste collection problems. For this reason, they have been considered in many vehicle routing problems applied to several areas. Nevertheless, in waste collection studies they have often been neglected.

Maximum shift duration constraints are rather straightforward, being used to ensure that each route is no longer than the duration of collection team shifts. This kind of constraints has been included in different vehicle routing problems and for different applications. Here with we refer some examples. [Caramia and Guerriero \(2010\)](#) used maximum shift duration (route time) constraints in a routing problem for milk collection. [Li et al. \(2016a, 2016b\)](#) considered similar constraints in a model formulation to address a routing problem in linehaul-delivery systems. [Mendoza et al. \(2016\)](#) proposed its introduction in a routing problem for servicing customers with uncertain demands through two approaches: establishing a probability of exceeding the maximum shift duration to be lower than a given threshold, and introducing penalties in the objective function for violations of the duration constraints. [Stavropoulou et al. \(2019\)](#) included maximum shift duration constraints in a VRPP model to optimize visits to customers. Considering waste collection vehicle routing problems, these constraints appeared in a couple of papers. [Teixeira et al. \(2004\)](#) used a combination of heuristics for planning vehicle routes on a monthly basis to collect recyclable urban waste in the center-littoral region of Portugal, considering maximum shift duration constraints. [Fanti et al. \(2016\)](#) considered this type of constraints in a waste collection VRP formulated as a mixed integer linear programming problem and solved it using heuristics. This approach was applied to a one-day waste collection case study of an Italian city.

Other workload concerns relate to route balance, specifically to the distribution of effort across collection teams. These teams need to be treated equitably, i.e., there should be no teams working many hours, while other ones work just a few and are not busy during their whole shift or most of its duration. The duration of a route can be quantified using different measures, such as route travelled distance, route time (which, depending on the problem, may include travel time, service time and waiting time), and route load. The two main objectives proposed in the literature to address route workload balance issues are: the minimization of the difference between the longest route and the shortest route, and the minimization of the longest route duration. [De Santiago et al. \(2013\)](#) and [Sivaramkumar et al. \(2018\)](#) focused on the former objective using time to measure the length of routes. The same objective was considered by [Jozefowicz et al. \(2002, 2006, 2007\)](#), [Jozefowicz et al. \(2009\)](#), [Martínez-Salazar et al. \(2014\)](#) and [Lacomme et al. \(2015\)](#), but measuring length by the travel distance. [Reiter and Gutjahr \(2012\)](#), [Sarpong et al. \(2013\)](#), and [Sun et al. \(2017\)](#) proposed the minimization of the longest route length. [Zhang et al. \(2019\)](#) also addressed this objective in terms of what they called longest route. With respect to other ways of achieving route workload balance, papers by [Lee and Ueng \(1999\)](#) and [Halvorsen-Weare and Savelsbergh \(2016\)](#) are highlighted here. The former authors proposed the minimization of the total difference between each driver working time and the shortest working time across drivers. [Halvorsen-Weare and Savelsbergh \(2016\)](#) focused on the distance travelled to measure the route duration, proposing four

different ways of balancing the routes: the two previously mentioned, the minimization of the difference between each route length and a desired route length (pre-defined), and the minimization of the difference in the distance travelled between all individual routes and the mean travelling distance. The applications of the previous papers are not related to waste collection. Regarding waste collection, [Kim et al. \(2006\)](#) introduced route workload balance constraints in their model for a North American waste management company operating nearly 26,000 collection and transfer vehicles. The model does not take into consideration the decision on which bins are profitable to visit. More recently, other authors have also integrated these concerns in their approaches to solve waste collection VRPs (without profits). [López-Sánchez et al. \(2018\)](#) as well as [Delgado-Antequera et al. \(2020a\)](#) proposed the minimization of the duration of the longest route. [Delgado-Antequera et al. \(2020b\)](#) proposed two objectives for route balancing: the minimization of the longest route in terms of distance and the minimization of the difference between the longest and the shortest routes in terms of duration.

3.4. Hybrid metaheuristics in VRPs

Hybrid metaheuristics have been extensively used to solve problems of the vehicle routing family, such as the vehicle routing problem with time windows (VRPTW), the capacitated VRP (CVRP), the VRP with cross-docking, the VRP with simultaneous pickup and delivery (VRPSD), the multi-depot VRP, and the heterogeneous VRP (HVRP). Different metaheuristics have been combined for hybridization. [Bent and Van Hentenryck \(2004\)](#) proposed a two-stage hybrid metaheuristic for solving the VRPTW. In the first stage, their metaheuristic minimizes the number of vehicles using simulated annealing (SA). A large neighborhood search (LNS) is then used in the second stage to minimize travel cost by relocating a large number of customers in the routes. This method was applied to the Solomon benchmark instances ([Solomon, 1987](#)), matching or improving the best solutions in 99% of the cases. [Benjamin and Beasley \(2010\)](#) proposed a hybrid metaheuristic based on variable neighborhood tabu search to solve a waste collection VRPTW. The algorithm developed by these authors performed better than previous work presented in the literature for publicly available waste collection problems involving up to 2092 customers. [Küçükoglu and Öztürk \(2015\)](#) proposed the combination of SA and tabu search (TS) to solve a VRP with backhauls and time windows. This metaheuristic was applied to a problem set from Solomon's VRPTW instances with 25, 50 and 100 nodes, presenting good solutions within practicable runtimes. [Küçükoglu et al. \(2019\)](#) also proposed a metaheuristic combining SA and TS to solve an electric travelling salesperson problem with time windows and mixed charging rates. [Yassen et al. \(2017\)](#) proposed the combination of SA with a harmony search algorithm. This hybrid metaheuristic was also applied to the Solomon's VRPTW instances, achieving good results when compared to other methods. [Pan et al. \(2020\)](#) developed a hybrid adaptive large neighborhood (ALN) with TS to solve a time-dependent VRPTW taking advantage of the ALN's strength of diversification and the strength of intensification of the TS. Computational results demonstrated good performance on benchmark instances, being optimal solutions obtained for instances up to 25 customers and best-known solutions improved for 36 instances with 50 and 100 customers. Ant colony optimization (ACO) coupled with LNS was proposed by [Akpınar \(2016\)](#) to solve a CVRP, presenting satisfactory results. [Liu and Jiang \(2019\)](#) solved a CVRP with time windows (CVRPTW) by integrating a large neighborhood search (LNS) metaheuristic with a genetic algorithm (GA), proving the hybrid metaheuristic efficiency through existing benchmarks. [Baniamernian et al. \(2019\)](#) combined a variable neighborhood search (VNS) heuristic with a GA to solve a profitable heterogeneous VRP with cross-docking. This hybrid metaheuristic was tested on a set of real-world inspired instances, presenting good performance and significantly better results than other algorithms used to solve the same instances. [Kalayci and Kaya \(2016\)](#) hybridized

ACO and VNS to solve a VRPSPD and evaluated the performance of their metaheuristic on a number of benchmark instances. The authors said this metaheuristic combines the powerful intensive local search of the VNS with the memory structure of the ACO. They concluded that the hybrid metaheuristic outperformed the performance of ACO or VNS when applied separately to the same instances. Soto et al. (2017) proposed a combination of multi-neighborhood search with TS to solve a multi-depot VRP. Tests on problem instances from the literature showed that this hybrid metaheuristic outperformed the state-of-the-art methods. Jabir et al. (2017) coupled ACO with VNS to solve a capacitated multi-depot VRP. These authors compared the solutions obtained through the metaheuristic with the solutions obtained through an exact algorithm, proving its efficiency, and with the solutions obtained by applying ACO alone, concluding that combining it with VNS improves the solutions. Delgado-Antequera et al. (2020b) developed a method composed by local search and pairwise path-relinking search to solve a CVRP for waste collection. The method was applied to a rural area of Spain, allowing significantly better results than the ones achieved by the waste collection company. In the same year, Delgado-Antequera et al. (2020a) proposed an iterated greedy algorithm coupled with VNS to solve a multi-objective VRP for waste collection. This approach was tested on a set of generated and real instances, and compared to other well-known methods from the literature, proving to be more effective in solving the problem.

3.5. Closing remarks

From the analysis of the aforementioned papers, it can be concluded that some progresses have been made on the study of VRPPs and the dynamic waste collection problem. However, the available approaches are still insufficient in terms of the size of the instances they can handle and due to the omission of important concerns in practice, such as the ones related to workload. Exploring and developing an approach that allows solving large dynamic waste collection VRPP while taking workload concerns into account is our main objective in this paper. Additionally, it can be concluded that hybridizing two metaheuristics in the same approach makes possible to take advantage of the strengths of each one of them, improving the final solutions. In this paper, simulated annealing (SA) is selected to investigate the solution space globally, and neighborhood search (NS) is subsequently applied to explore in detail the vicinity of the solution produced by simulated annealing, in order to guarantee that, at least, a local optimum is found.

4. Hybrid metaheuristic

The hybrid metaheuristic we propose is divided into two phases (see Fig. 1): Phase 1 consists of a procedure to determine whether collection is needed and, if so, the must-go bins; and Phase 2 consists of a hybrid simulated annealing and neighborhood search (SANS) algorithm to define profitable routes for each collection day. Phase 1 input data are the bin fill levels retrieved by sensors for that day, but also historical bin accumulation rates for the next days (look-ahead vision). Phase 2 input data are the must-go bins identified in Phase 1 and the bin fill levels of all the bins. The constraints that need to be respected are related to bin capacities, vehicle capacities, and maximum shift duration. The objective of this smart waste collection problem is to maximize the profit of the collection company and the route workload balance. The mathematical formulation of this problem (without workload concerns) can be found in Ramos et al. (2018a). As mentioned, Fig. 1 describes the structure of the hybrid metaheuristic. Its two phases are explained in detail in the remainder of this section.

4.1. Phase 1 (collection days and must-go bins)

As stated previously, the collection days and must-go bins for each collection day are determined in Phase 1 of our metaheuristic. The

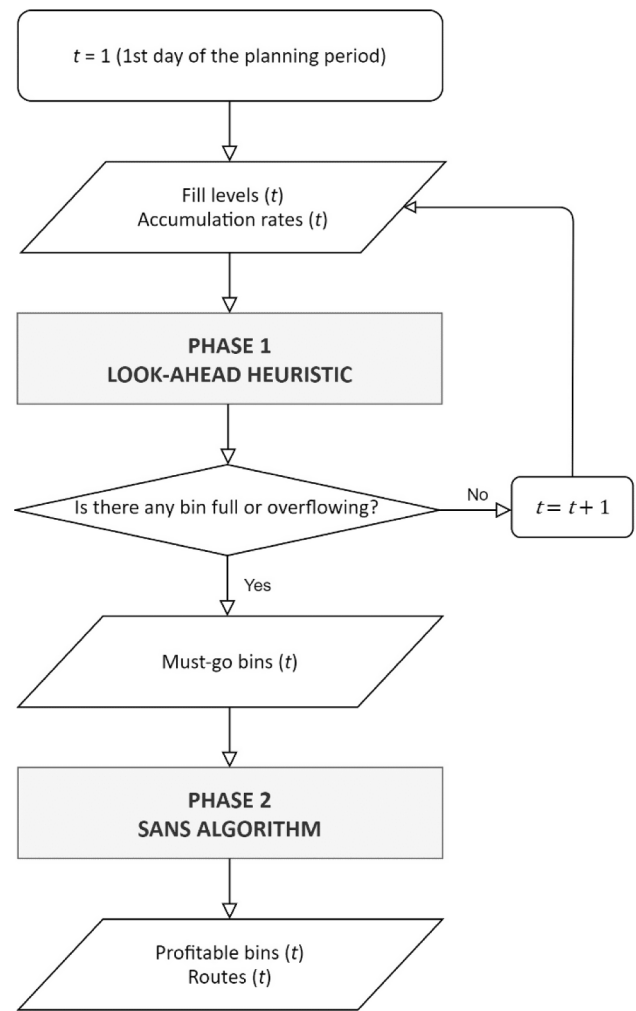


Fig. 1. Structure of the hybrid metaheuristic.

definition of the collection days is based on the existence (or not) of at least one bin completely full or overflowing ($W_{it} \geq E_i, \forall i \in I, t \in T$), where W_{it} is the stock in bin i at day t and E_i is the capacity of bin i . Full or overflowing bins are must-go bins for each collection day. In addition, bins that are estimated to become full between the first collection day and the following collection day are also must-go bins. The following collection day is the day in which the bins that were full or overflowing in the first collection day are predicted to become full again. This allows looking further in the planning period (look-ahead vision), avoiding unnecessary routes between each collection day and the following estimated collection day. Historical data is used to estimate the state of the bins in terms of fill levels for each day in the future. Inspired by the literature on inventory management (Waters, 2013), bin fill levels are estimated by considering the average daily accumulation rate ($\bar{a}_i$) and the standard deviation of this rate (σ_{ai}) to capture the uncertainty associated to it (assuming a normal distribution) through the following expression:

$$\hat{W}_{it'} = n_{it'} \times \bar{a}_i + \alpha \times \sqrt{n_{it'}} \times \sigma_{ai}, \forall i \in B, t' \in [t+1, T] \quad (1)$$

where

$\hat{W}_{it'}$: estimated waste stock in bin i at day t' ;

$n_{it'}$: number of days between collection day t and day t' ;

$\bar{a}_i$: average daily accumulation rate of bin i ;

α : value for a probability retrieved from the normal distribution

depending on the desired level of confidence for $\hat{W}_{it'}$; and

σ_{ai} : standard deviation of the daily accumulation rates of bin i .

A flowchart of the procedure performed in this phase is presented in Fig. 2.

To clarify how this procedure works, we present next a simple case involving 10 bins and 5 days (Table 1). For each day after the first collection day (day 1), the waste stock at each bin before and after collection on day 1 is displayed in different columns in Table 1. The capacity of the bins is 2.5 m^3 , which corresponds to a weight of 74 kg considering a reference waste density of 29.5 kg/m^3 . Since, on day 1, bins #1 and #5 are, respectively, full and overflowing, a collection route must be performed that day. Given the average accumulation rates of bins #1 and #5, the next time one of them is expected to be full or overflowing again is day 5. Indeed, in this day, $\hat{W}_{55}$ is greater than 74 kg. To avoid this, day 4 should be the following collection day. Between day 1 and day 4, only bins #7 and #9 are expected to overflow, so these are the bins that join #1 and #5 in the set of must-go bins on day 1. Summarizing, bins #1, #5, #7 and #9 are must-go bins on day 1, and the decision on profitable bins to collect (made on Phase 2 of the metaheuristic) focuses only on bins #2, #3, #4, #6, #8 and #10. After the collection route on day 1 is performed, this procedure is carried out again on day 2 with the bin fill levels retrieved by the sensors. If any bin is full or overflowing, a new collection route must take place this day. Otherwise, no route needs to be performed. This procedure is repeated every day to check whether a collection route is necessary that day and, if so, the must-go bins.

4.2. Phase 2 (profitable route selection)

For each collection day and considering the must-go bins defined in Phase 1, a hybrid simulated annealing/neighborhood search (SANS) algorithm is used in Phase 2 to select the profitable bins to collect and the respective collection routes. The structure of this algorithm is represented in Fig. 3. It is composed by two stages – simulated annealing and neighborhood search – which are intercalated with a procedure to eliminate intersections between arcs in routes. These stages are characterized in detail in the next subsections.

4.2.1. Simulated annealing algorithm

Simulated annealing (SA) is a stochastic global search method developed to solve combinatorial optimization problems as the one under consideration in this paper. SA was chosen to integrate our hybrid metaheuristic because this type of algorithm generally performs well in routing problems and is easy to code (in the paper it was introduced – Kirkpatrick et al., 1983 – the merits of SA were illustrated using precisely the traveling salesperson problem as example). It is inspired by the process of annealing in metallurgy, which involves the heating and controlled cooling of a metal. In simulated annealing, this is interpreted as a slow decrease in the probability of accepting worse solutions as the solution space is explored, i.e., accepting worse solutions is frequent at the beginning to allow a more extensive search for the global optimal solution. When the end is approaching, this becomes less and less frequent, and the probability of accepting worse solutions is almost zero.

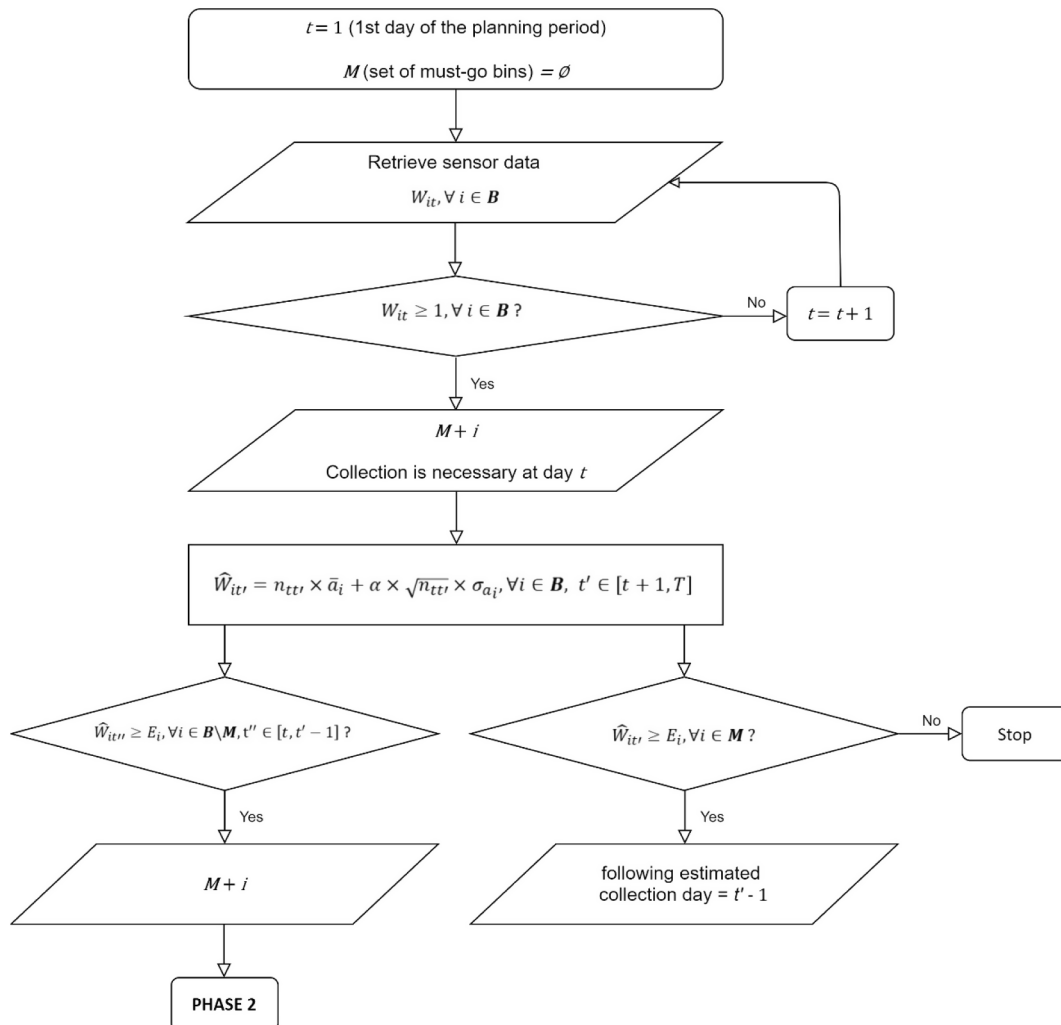


Fig. 2. Phase 1 heuristic.

Table 1
Simple case illustrating the Phase 1 heuristic.

Bin	Expected waste accumulation(kg/day)	Day 1 Waste stock retrieved from sensors (kg) $\hat{W}_{it}$	$\hat{W}_{it}$ Estimated waste stock (kg)							
			Day 2		Day 3		Day 4		Day 5	
			Before collection	After collection	Before collection	After collection	Before collection	After collection	Before collection	After collection
1	10	74	84	10	94	20	104	30	114	40
2	10	10	20	20	30	30	40	40	50	50
3	2	55	57	57	59	59	61	61	63	63
4	5	47	52	52	57	57	62	62	67	67
5	20	85	105	20	125	40	145	0	165	20
6	4	36	40	40	44	44	48	48	52	52
7	10	70	80	10	90	20	100	30	110	40
8	5	40	45	45	50	50	55	55	60	60
9	5	65	70	5	75	10	80	15	85	20
10	2	43	45	45	47	47	49	49	51	51

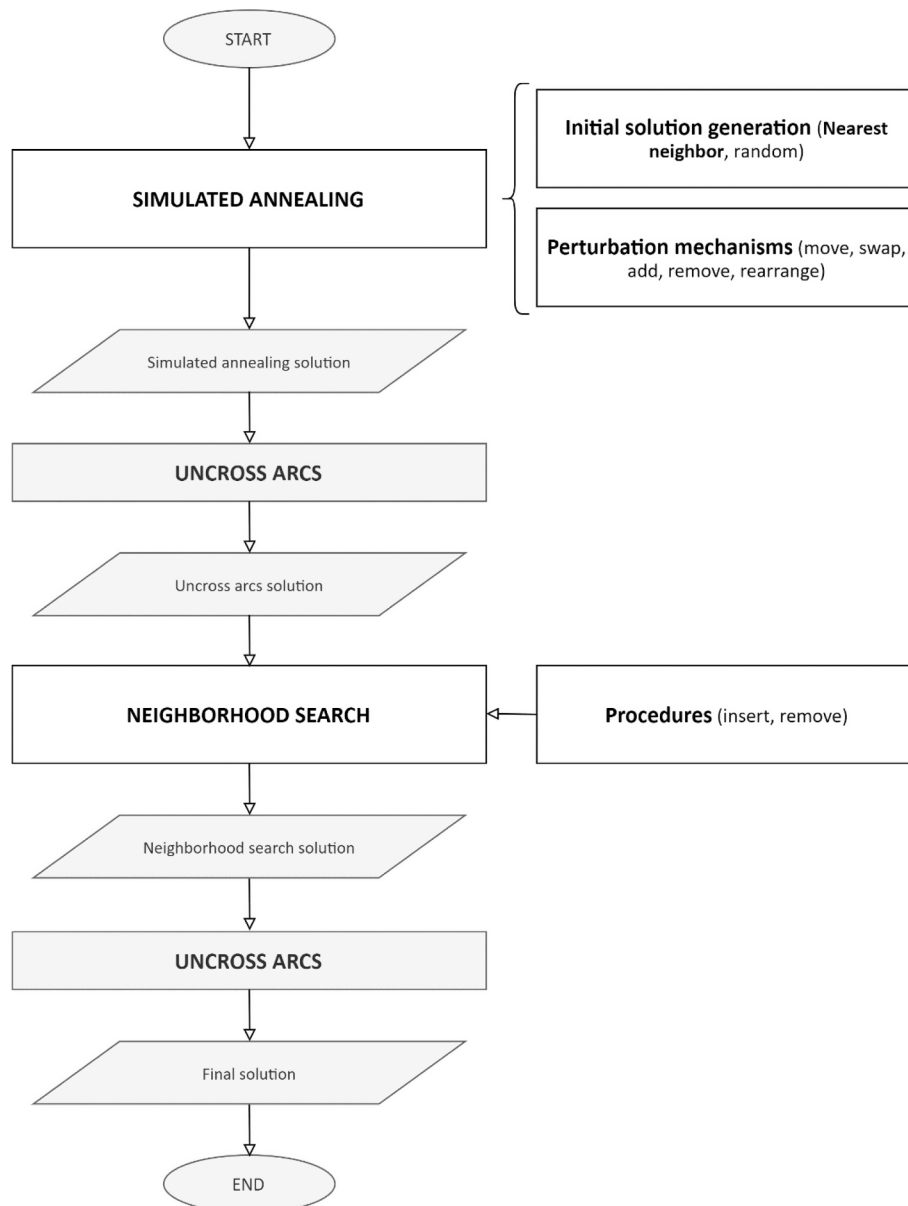


Fig. 3. Structure of the simulated annealing/neighborhood search algorithm.

SA starts with an initial solution, set to Sol_{cur} , and an initial temperature (which is a SA parameter) used to control the probability of accepting a worse solution at iteration 1. In each iteration, a solution (Sol_{new}) derived from the current solution (Sol_{cur}) by using a perturbation mechanism is created and evaluated according to an objective function. If Sol_{new} is better than or equal to Sol_{cur} , Sol_{new} becomes Sol_{cur} ; if it is worse, it may become Sol_{cur} with a probability. This probability (associated to the temperature) progressively decreases from an initial positive value to zero. Several iterations are performed until a stop condition is met. Therefore, there are several aspects to take into consideration in order to apply simulated annealing, namely:

- Setting an initial solution;
- Formulating the objective function;
- Choosing a cooling function (or schedule);
- Selecting the perturbation mechanisms to change the solution in each iteration;
- Defining the stop condition.

5. Initial solution

The initial solution (i.e., the initial route or set of routes) includes all

bins. Two classic procedures were tested for sequencing bins in initial route(s): nearest neighbor and random choice. The nearest neighbor procedure is well known and consists of starting from a node and successively adding the closest node to the previous node. In the case of a waste collection problem, a route is initialized at the depot (where vehicles are parked), the closest bin to the depot that is not yet in any route is added, the closest bin to the previous bin that is not in any route yet is added, and so forth. Following this process, bins are added to a route until either a vehicle capacity or a maximum shift duration constraint is violated. When this happens, a new route is created. The procedure stops when all bins are part of the initial solution. The random choice procedure is similar, but random bins are selected instead of closest bins. It was concluded through some experiments that the nearest neighbor procedure, described in Pseudo-code 1, performs better than the random choice procedure. Therefore, all initial solutions generated in our study are based on the nearest neighbor procedure. Our experiments also suggested that it is advantageous to divide bins in sets according to their location (longitude), to avoid letting bins behind and, consequently, having to sequence two bins very far from each other. The procedure only jumps to the closest bin in a different set when all the bins in the current set are visited.

Pseudo-code 1. Initial solution generation procedure.

```

 $B_1, B_2, B_3$ : sets of bins in areas 1, 2 and 3, respectively
 $H$ : depot
 $Q$ : vehicle capacity
 $S_{max}$ : maximum shift duration
 $o$ : bin collection time
 $IS$ : initial solution

Set  $IS = \emptyset$ 

While ( $B_1 \neq \emptyset$  and  $B_2 \neq \emptyset$  and  $B_3 \neq \emptyset$ )
  Set  $route = []$ 
  Set  $route_{load} = 0$ 
  Set  $route_{duration} = 0$ 
  Add  $H$  to  $route$ 
  Set  $prev_{bin} = H$ 
  For  $i$  in [1, 3]:
    While( $B_i \neq \emptyset$  and  $route_{load} < Q$  and  $route_{duration} < S_{max}$ ):
      Set  $cur_{bin} = \text{closest bin to the previous one that is not in any route yet}$ 
      Set  $route_{load} = route_{load} + stock_{cur_{bin}}$ 
      Set  $route_{duration} = route_{duration} + traveltime_{cur_{bin}, prev_{bin}} + o_{cur_{bin}}$ 
      If  $route_{load} < Q$  and  $route_{duration} < S_{max}$ :
        Add  $cur_{bin}$  to  $route$ 
      End if
    End while
  End for

  Add  $H$  to  $route$ 
  Add  $route$  to  $IS$ 
End while

```

5.1. Objective function

The reference objective function considered is the one proposed by Ramos et al. (2018), in which the profit is maximized. The profit corresponds to the difference between the revenue obtained by selling the collected waste and the cost of collecting the waste. The maximization of route workload balance is added to the objective function proposed by Ramos et al. (2018) through the use of a penalty in order to convert the multi-objective problem into a single-objective one. Moreover, we introduce penalties in the objective function to handle vehicle capacity and maximum shift duration constraints in the solution approach. These penalties are calibrated in order to prevent vehicle capacity and maximum shift duration to be exceeded.

The notation used to formulate the objective function is as follows:

Sets

$B = \{1, \dots, B\}$: set of bins, indexed by i

$T = \{1, \dots, T\}$: set of days in the planning period, indexed by d

Decision variables

y_i : binary variable equal to 1 if bin i is collected and equal to 0 otherwise

x_{ij} : binary variable equal to 1 if arc (i, j) is visited and equal to 0 otherwise

z_{max} : duration of the longest route

z_{min} : duration of the shortest route

Parameters

r : revenue obtained by selling the collected waste

c : travel cost by unit travelled

Q : vehicle capacity

W_{it} : waste stock in bin i at day t

d_{ij} : distance between bin i and bin j

P_Q : penalty associated to vehicle capacity violations

P_S : penalty associated to maximum shift duration violations

o : time spent to collect each bin

$\bar{s}$: average travel speed

P_Z : penalty associated to route workload balance differences

S_{max} : maximum shift duration

Using this notation, the objective function can be formulated as follows:

$$MaxProfit = r \times RT - (c \times TCT + P_Q \times LT + P_S \times SDT + P_Z \times RBT) \quad (2)$$

$$RT = \sum_{i \in N, i \neq Depot} W_{it} \times y_i \quad (3)$$

$$TCT = \sum_{i \in N} \sum_{j \in N, i \neq j} x_{ij} \times d_{ij} \quad (4)$$

$$LT = \left(\sum_{i \in N} W_{it} \times y_i - Q \right) \quad (5)$$

$$SDT = \left(o \times \sum_{i \in N} y_i + \frac{\left(\sum_{i \in N} \sum_{j \in N, i \neq j} x_{ij} \times d_{ij} \right)}{\bar{s}} \right) - S_{max} \quad (6)$$

$$RBT = z_{max} - z_{min} \quad (7)$$

The objective function (Eq. (2)) is divided into five terms (equations (3), 4, 5, 6, and 7), respectively related to revenue (RT), travel cost (TCT), and the three penalties (vehicle capacity (LT), maximum shift duration (SDT), and route workload balance (RBT)). As it can be seen, route workload balance is evaluated based on the difference between the duration of the longest and the shortest routes. The duration includes the stop time to collect each bin in the route plus the travel time.

5.2. Cooling function

The chosen cooling function is the Boltzmann distribution usually used in SA algorithms. This function is as follows:

$$p_i = e^{\Delta S_i / T_i} \quad (8)$$

where:

p_i : probability of accepting solution of iteration i (Sol_{new});

$\Delta S_i = Prf_{new} - Prf_{cur}$: difference between Prf_{new} (profit associated to Sol_{new} – iteration: i) and Prf_{cur} (profit associated to Sol_{cur} – previous iteration: $i-1$);

$T_i = T_0 / i^\theta$: temperature at iteration i ;

T_0 : initial temperature (iteration 1);

θ : parameter to reduce the decreasing rate of the temperature. The lower the θ , the lower the decreasing rate.

5.3. Perturbation procedures

Several perturbation procedures are developed to generate new solutions, Sol_{new} namely: *move swap remove add and rearrange*. Details on these procedures can be seen in Appendix B.

5.4. Stop condition

With respect to the stop condition, there are different possible options, such as:

- Runtime – the SA algorithm stops when the maximum runtime is reached;
- Number of iterations – the SA algorithm stops when the maximum number of iterations is reached;
- Number of iterations without any solution improvement – the SA algorithm stops if, in a given number of consecutive iterations, the Sol_{cur} does not improve.

In our metaheuristic, two conditions are chosen depending on the size of the problem to solve. For larger problems, the determinant factor may be the runtime, and therefore this is the condition used. If problems are smaller, there are no runtime issues and the condition chosen is the number of iterations.

Considering the above aspects, the SA algorithm is summarized in Pseudo-code 2.

Pseudo-code 2. Simulated annealing algorithm.

5.4.1. Neighborhood search algorithm

As depicted in Fig. 3, the NS algorithm is performed after the SA. It aims at further exploring the neighborhood of the SA solution in order to find better solutions in this neighborhood and ensure that, at least, a local optimum is found. The starting point is the solution obtained from the SA metaheuristic. The neighborhood is explored by considering insertion and removal procedures.

Three different procedures are considered: a *remove* procedure and two different *insert* procedures (A and B). Details on these procedures can be found in Appendix C.

The number of times the neighborhood search procedures are repeated depends on the improvements in profit or on the runtime. If the runtime is less than the 2 h, the algorithm stops or moves to the next procedure when, in two successive runs of a procedure, there is no improvement. Otherwise, the procedures are performed until the runtime is reached.

5.4.2. Uncross arcs procedure

During the execution of the SA and the NS algorithms, it is possible that the solutions found contain routes with arc intersections as the one illustrated in Fig. 4 on the left. This is clearly more inefficient in terms of

```

Set  $T_0$ 
Set  $\theta$ 
Define set of possible perturbation mechanisms
Generate  $Sol_{cur}$  from  $IS$ 
Calculate  $Prf_{cur}$ 
While (stop condition)
    Randomly choose perturbation mechanism ( $PM$ ) from set of
    possible perturbation mechanisms
    Generate  $Sol_{new}$  by applying  $PM$  to  $Sol_{cur}$ 
    Calculate  $Prf_{new}$ 
    Calculate  $\Delta S$ 
    If  $\Delta S \geq 0$  then
        Accept  $Sol_{new}$ 
         $Sol_{cur} = Sol_{new}$ 
         $Prf_{cur} = Prf_{new}$ 
    Else
        Calculate  $T$ 
        Calculate  $p(Sol_{new})$ 
        If  $p(Sol_{new}) \geq random(0,1)$ 
            Accept  $Sol_{new}$ 
             $Sol_{cur} = Sol_{new}$ 
             $Prf_{cur} = Prf_{new}$ 
        End if
    End if
End while
 $Sol_{SA} = Sol_{cur}$ 
 $Prf_{SA} = Prf_{cur}$ 

```

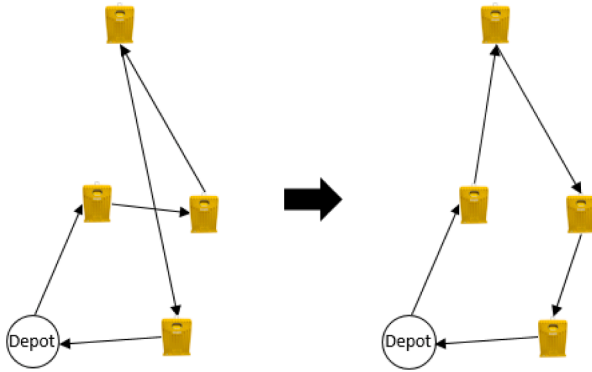


Fig. 4. Uncross arcs procedure.

profit than the route on the right side. To deal with such occurrences, an uncross arcs procedure is developed that uncross all the existing arc intersections in each route, allowing to convert solutions as the one in Fig. 4 on the left to solutions as the one in Fig. 4 on the right. This procedure is run after the termination of the SA algorithm and after each iteration of the NS algorithm, as shown in Fig. 3.

5.5. Calibration of metaheuristic parameters

The hybrid metaheuristic involves the following parameters: α (collection days and must-go bins selection heuristic), T_0 , θ and N (number of iterations – SA algorithm), P_Q , P_S , and P_Z (penalties in the objective function – SA and NS algorithms), and N_A (NS algorithm –

insertion procedure A). The maximum runtime (R_{max}) is also an important parameter for SA and NS algorithms when larger instances are at stake.

For calibrating these parameters excluding α and R_{max} , an interval of variation was defined for each parameter as well as a step according to which the value of the parameter could vary inside the interval of variation. With respect to α and R_{max} , the values considered were:

α : 0.52 or 0.84 (normal distribution level of confidence of 70% or 80%, respectively)

R_{max} : 2 h (as previously said, only 2 h are available to run the model in practice, so this is the only possibility when the runtime is the stop condition)

Multiple combinations of possible values of the parameters were generated and tested using randomly-generated instances (described in Section 5). The combination with the best performance, evaluated by the objective function value, was the chosen one. The intervals of variation, steps, and best values for the different parameters can be found in Table 2.

Table 2
Metaheuristic parameters calibration.

Parameter	Interval of variation	Step	Best value
α	–	–	0.52
T_0	[50, 200]	25	75
θ	[0.5, 2]	0.1	0.7
N	[5000, 80000]	5000	See Table 3
P_Q	[0.080, 0.1]	0.005	0.095
P_S	[0.1, 1]	0.1	0.5
P_Z	[0.1, 1]	0.1	0.5
N_A	[10, 50]	10	50

The best combination of parameters $\alpha, T_0, \theta, P_Q, P_S, P_Z$, and N_A was the same for all test instances. It should be noted that when workload concerns are not considered, P_S and P_Z should be 0 instead of 0.5.

As regards parameter N , its best values was found to depend on instance size (number of bins, B), as well as on the relationship between W_{tot} (total weight in all bins retrieved by the sensors when the metaheuristic is triggered) and the maximum weight if all bins were full ($W_{max} = B \times Q$) – see the total weight of waste for each instance in [Appendix D](#). For instances with 250 bins or less, N was the determinant stop condition varying according to [Table 3](#).

With respect to instances with 500 bins, the determinant stop condition was given by R_{max} .

6. Performance analysis

This section aims at analyzing the performance of the SANS algorithm (Phase 2) of the proposed hybrid metaheuristic. It is composed by three subsections. In the first one, the test instances generated to perform this analysis are described as well as the data used. The second subsection focuses on the performance of the SANS algorithm without considering workload concerns. More specifically, it presents: a comparison between the solutions found by the SANS algorithm and the solutions found by the exact branch-and-cut algorithm (GAMS-CPLEX) regarding quality and runtimes; the performance of the SANS algorithm for different runtimes; and the role played by each of the algorithms (SA and NS) in the solution search process. The last subsection is dedicated to the performance of the SANS algorithm when workload concerns are included and consists of: a comparison between the results of the SANS algorithm with and without workload concerns; and again, an analysis of the role played by SA and NS in the search process.

6.1. Test instances

For analyzing the performance of the hybrid metaheuristic, we considered 25 randomly-generated instances of different sizes (i.e., number of bins): 10 instances of 50 bins, 5 of 100 bins, 5 of 250 bins, and 5 of 500 bins. The data for these instances are available as [supplementary materials](#). Specifically, we defined at random the location (latitude and longitude) of the depot and the bins within a given region (equal for all instances), as well as the waste stock on a given day and the accumulation rate for each bin. According to fill level, bins are divided into five intervals: between 0% and 25%, between 25% and 50%, between 50% and 75%, between 75% and 100%, and equal to 100% or more. The must-go bins are the ones that are full (100%), already overflowing or estimated to overflow in the next day (based on the accumulation rate). Examples of instances considered are shown in [Fig. 5](#). In this figure, bins are represented by different colors according to their fill levels (dark green corresponds to the bins with less waste and red corresponds to full or overflowing bins). The total weight of waste in bins transmitted by the sensors when the metaheuristic is triggered (W_{tot}), and the number of bins per fill level interval for each instance are provided in [Appendix D](#).

The data for these instances consist of: B, V, H , where each route starts and ends, T, d_{ij} for each edge (i, j) in the undirected graph defined on $B + 1, E_i, Q, S_{max}, c, r, o$, and $\bar{s}$. All of the values used in the test instances are the ones provided by Valorsul to the real-world study

Table 3
Number of iterations for instances with less than 500 bins.

B	W_{tot}	N
50	less than $0.6 \times W_{max}$	5000
50	$\geq 0.6 \times W_{max}$	50,000
100	less than $0.55 \times W_{max}$	5000
100	$\geq 0.55 \times W_{max}$ and $\leq 0.65 \times W_{max}$	50,000
100	$0.65 \times W_{max}$	80,000
250	Any	80,000

presented in [Section 6](#). The travel cost per unit of distance includes fuel consumption, vehicle maintenance and collection team wages. The revenue is calculated considering the price paid to Valorsul by Sociedade Ponto Verde (the packaging waste regulator in Portugal) for each ton of paper/cardboard collected and sorted by a waste collection company. This price was 136 €/ton in the planning period considered here. As only the collection is considered in this paper, this value needs to be reduced. According to [Ramos et al. \(2014a, 2014b\)](#), 70% of the price corresponds to the collection, resulting in 95.2 €/ton. The amount (weight) of waste collected from each bin i at each day t , W_{it} , registered by the collection team (considering the density of paper/cardboard $\rho = 29.5 \text{ kg/m}^3$) was assumed to be the amount of waste that sensors would read and transmit every day. The values for the scalar parameters of the test instances are presented in [Table 4](#).

6.2. SANS algorithm without workload concerns

Using all test instances, the performance of the SANS algorithm regarding solution quality and runtime was evaluated against the results obtained by the exact optimization algorithm used to solve the model proposed by [Ramos et al. \(2018a\)](#). Since this model does not include workload concerns, for comparison purposes, we ran the SANS algorithm without considering such concerns. The SANS algorithm has a strong random component associated to the perturbation mechanisms, leading frequently to different solutions to the same instance. Therefore, ten runs of the SANS algorithm were performed for each test instance in order to account for the effects of the randomness in the analysis/comparison. In [Table 5](#), the results obtained through the exact algorithm and the SANS algorithm (best run) are compared. It should be noted that the runtime limit was set to five hours (18,000 s) in the case of the exact method and to two hours (7200 s) in the case of the hybrid metaheuristic.

Regarding solution quality (profit), [Table 5](#) shows that the SANS algorithm is effective in finding good-quality solutions. Regarding 50-bin instances, hybrid metaheuristic solutions are just 0.7%, on average, far from the optimal solutions, being the optimal solution obtained for 7 out of the 10 instances. For the instances with 100 bins, results obtained through the SANS algorithm are, on average, only 3.5% far from the optimal results. In these size-instances, the hybrid metaheuristic is able to find the optimal solution for 2 out of the 5 instances. The SANS algorithm is able to reach the optimum for some runs (only the best one is shown in the table, see [Appendix E](#) for the remaining ones). The performance of the SANS algorithm in terms of profit for 100-bin instances is sometimes not as good as its performance for instances of 50-bin instances. For 50-bin instances, the number of possible combinations of bins in routes after removing the bins not considered to be profitable is still relatively low and most of them are tried by the algorithm. This is not always the case for 100-bin instances. The locations of the depot, the must-go bins, and the bins with higher fill levels influence the performance (profit) of the SANS algorithm. For example, when comparing results of 100-bin instances 1, 2 and 3 with 100-bin instance 4 (see [Appendix E](#)), the performance of the SANS algorithm is much poorer for instance 4. This is due to the location of the depot regarding the must-go bins and the location of the bins with higher fill levels regarding the must-go bins. [Fig. 6](#) shows that there is a large percentage of bins with fill levels equal or higher than 50% far from the depot and far from the must-go bins (more on the right) for instance 4, which does not happen in the other three instances. This leads the SANS algorithm to decide not visiting these bins even if they are profitable.

Regarding runtimes, the exact optimization algorithm was much faster for the smallest instances (50 bins). For 100 bin instances, exact algorithm runtime was, on average, 9 times higher than the SANS algorithm runtime, but still under the maximum acceptable time in practice (2 h runs). This indicates that for instances of this size, there is no need to use the SANS algorithm.

For 250-bin instances (bins), the exact algorithm was not able to find

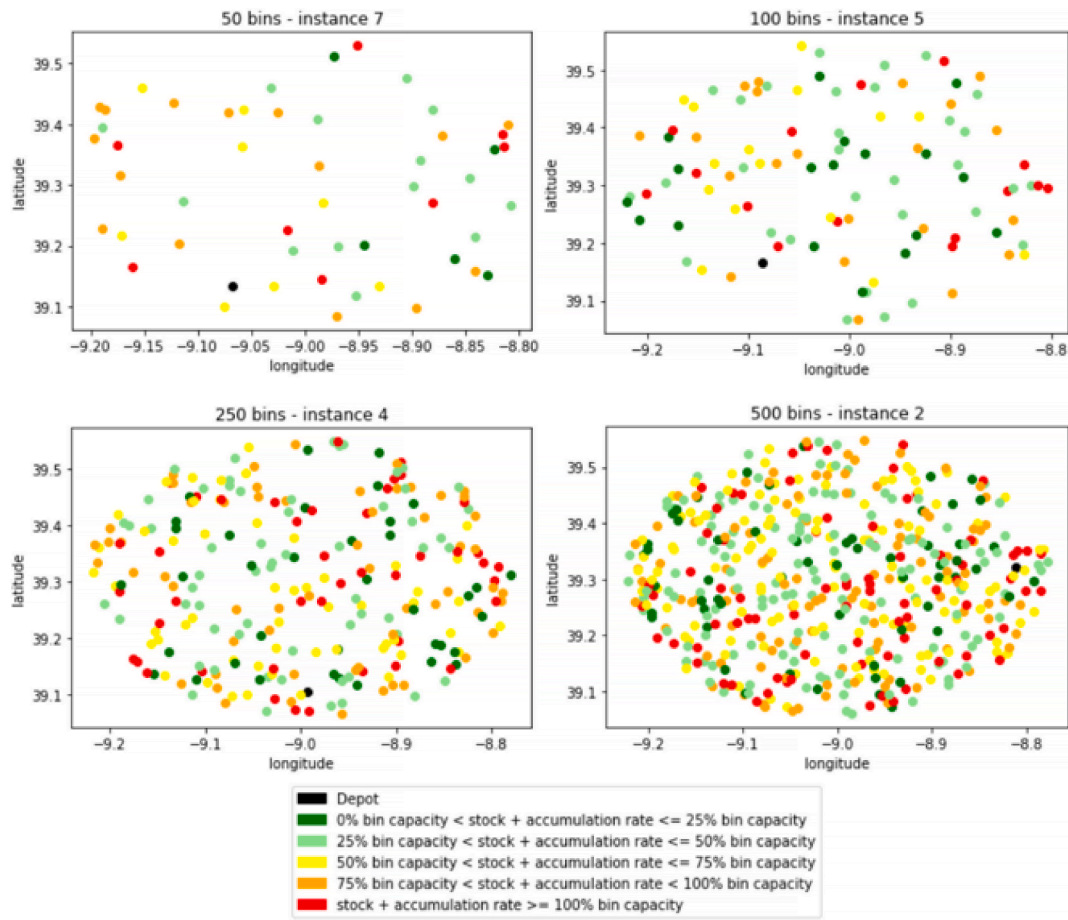


Fig. 5. Examples of test instances (50, 100, 250 and 500 bins).

Table 4
Scalar parameters of the test instances.

Parameter	Value
c	1 €/km
r	0.0952 €/kg
$S_{\max}$	4000 kg
Q	420 min
ρ	29.5 kg/m ³
o	3 min
S	40 km/h
E_i	74 kg
B	50 / 100 / 250 / 500 bins

any solution for instances 1 and 2 within 2 h (i.e., the runtime that is compatible with companies' operational constraints). Regarding instances 4 and 5, the solutions found by the SANS algorithm were significantly better than the solutions found by the exact optimization algorithm within 2 h. Only in the case of instance 3, the exact optimization algorithm performed better than the SANS algorithm within two hours. Considering the 2-hour computational time limit, the hybrid metaheuristic reaches profits that are, on average, 57% higher than the ones obtained by the exact method, being the average runtime slightly less than half of the runtime limit. If more time would be available to run the exact algorithm, the SANS algorithm solutions would be 11%, on average, far from the best solutions found by the exact algorithm. However, for instance 2, even considering 5 h of runtime, it is impossible for the exact optimization algorithm to find any solution. This demonstrates the limitations of using exact optimization to find solutions for larger problems, even when some important concerns (namely the ones related to workload) are excluded.

For the largest instances we tested (500 bins), the exact algorithm was not able to find any solution. On the contrary, the SANS algorithm always found solutions for these instances within the two-hour runtime limit. Considering the 10 runs, the difference between the average solution and the best solution and the difference between the worst solution and the average solution found by the metaheuristic were always under 5%, indicating good results stability. For these 500-bin instances, the runtime was a limitation. Therefore, two experiments were carried out considering the runtimes of two and four hours, respectively. Four-hour experiments were considered in order to analyze how the solutions could be improved if there were more time available in practice to run the SANS algorithm. It was concluded that running the SANS algorithm for longer time improves the results in general. However, profit increase was low (the maximum increase was 4.1% for instance 3, see Table 6).

In conclusion, the SANS algorithm proved to be computationally effective in dealing with larger size instances (250 and 500 bins), while simultaneously presenting good results. As the instance size increases, exact optimization runtime significantly increases, making its use prohibitive in practice. The SANS algorithm runtime also increases with the size of the instances, but at a much smaller pace, which enables its implementation in practice. More detailed results regarding the different instances and runs of the SANS algorithm are presented in Appendix E.

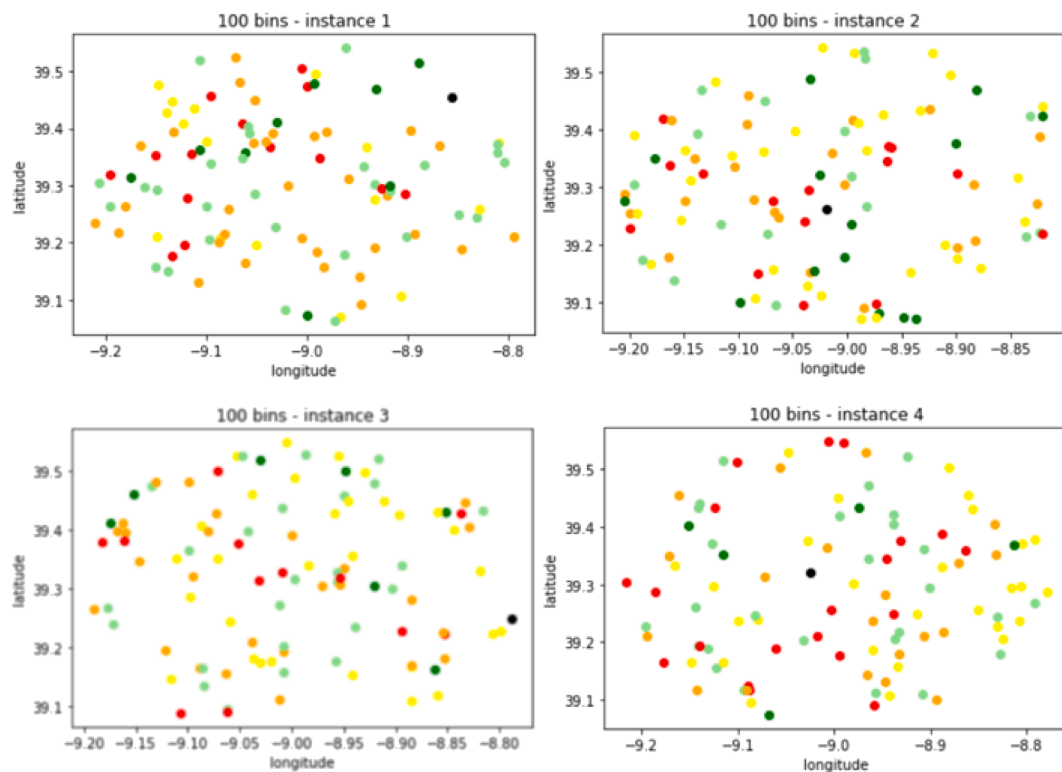
As previously described, the SANS algorithm is composed by two algorithms (SA and NS). The NS runs after the SA, taking as input the SA solution. It is important to analyze the role of each of the algorithms in the solution search process. We performed this analysis and concluded that the performance of the SA and NS algorithms was different depending on the instance size (number of bins). Indeed, it was verified that for instances with 50 and 100 bins, SA allows greater profit

Table 5

Comparison of SANS algorithm solutions with the exact algorithm solutions.

Instance		Exact algorithm (GAMS-CPLEX)					SANS algorithm		Deviation	
Size	Number	Upper bound (€)	Lower bound (€)	Runtime (s)	Profit (€)	Runtime (s)	Profit (€)	Runtime (s)	Profit	Runtime
50 bins	1	-69.37	-69.37	64	-	-	-71.26	38	-2.7%	-41%
	2	-22.36	-22.36	29	-	-	-22.36	36	0.0%	23%
	3	-41.60	-41.60	69	-	-	-41.6	182	0.0%	162%
	4	-58.25	-58.25	29	-	-	-58.25	55	0.0%	92%
	5	-28.06	-28.06	20	-	-	-28.46	36	-1.4%	80%
	6	-48.81	-48.81	28	-	-	-48.81	215	0.0%	677%
	7	-45.39	-45.39	15	-	-	-45.39	241	0.0%	1490%
	8	-78.57	-78.57	38	-	-	-78.57	67	0.0%	79%
	9	-43.74	-43.74	19	-	-	-43.74	211	0.0%	1002%
	10	-62.94	-62.94	50	-	-	-64.87	67	-3.1%	35%
	Average								-0.7%	359.9%
100 bins	1	39.19	39.19	4789	-	-	34.54	486	-11.9%	-90%
	2	56.74	56.74	5514	-	-	54.11	729	-4.6%	-87%
	3	45.27	45.27	8584	-	-	44.71	1107	-1.2%	-87%
	4	24.93	24.93	6344	-	-	24.93	910	0.0%	-86%
	5	19.23	19.23	6288	-	-	19.23	82	0.0%	-99%
	Average								-3.5%	-89.6%
250 bins	1	406.03	403.78	18,000	-	7200	344.53	3152	-14.7%	-82%
	2	389.36	-	18,000	-	7200	281.29	3209	-	-82%
	3	391.16	378.83	18,000	357.80	7200	318.28	2949	-16.0%	-84%
	4	399.14	359.89	18,000	-467.42	7200	349.88	2893	-2.8%	-84%
	5	419.29	403.23	18,000	333.10	7200	361.36	3120	-10.4%	-83%
	Average								-11.0%	-83.0%
500 bins	1	1034.28	-	18,000	-	7200	908.44	7200	-	-
	2	1031.72	-	18,000	-	7200	718.52	7200	-	-
	3	1137.51	-	18,000	-	7200	830.76	7200	-	-
	4	1079.37	-	18,000	-	7200	803.77	7200	-	-
	5	1061.67	-	18,000	-	7200	864.66	7200	-	-
	Average								-	-

Note: Italic numbers mean that the optimality has not been proven.

**Fig. 6.** 100 bins test instances 1, 2, 3 and 4 – bins location and fill levels.

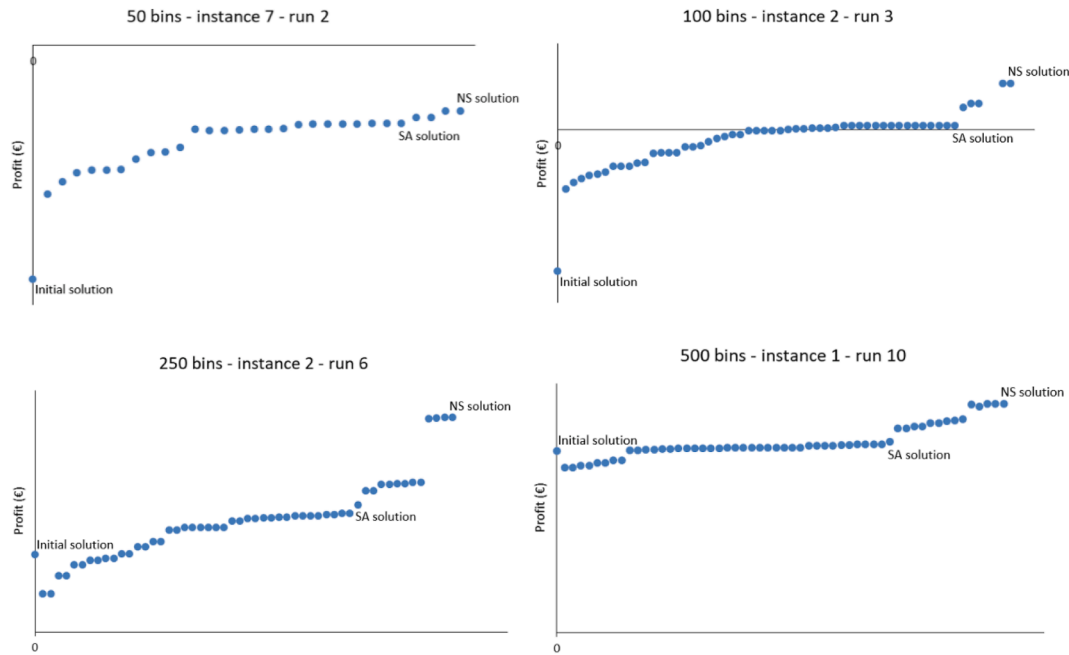
increases regarding the initial solution than NS regarding the SA solution. For 250-bins instances, the performances of SA and NS were very similar, being SA improvements slightly higher. Regarding 500-bin

instances, NS performs significantly better than SA. Fig. 7 illustrates this behavior with an example for 50, 100, 250 and 500 bins instances. Results thus indicate that, for larger instances, having a combination of

Table 6

Profit differences between 2- and 4-hour runtime experiments for 500-bin instances.

Instance	Profit (€) vs. Runtime								
	Highest profit			Lowest profit			Average profit		
	4 h-run	2 h-run	Difference (%)	4 h-run	2 h-run	Difference (%)	4 h-run	2 h-run	Difference (%)
1	925.78	919.77	0.6	904.08	879.51	2.7	915.65	908.44	0.8
2	751.10	743.57	1.0	703.31	702.25	0.2	728.07	718.52	1.3
3	880.31	843.82	4.1	806.92	811.68	-0.6	844.94	830.76	1.7
4	839.45	829.41	1.2	800.95	781.28	2.5	819.13	803.77	1.9
5	899.48	883.55	1.8	877.06	848.7	3.2	885.86	864.66	2.4

**Fig. 7.** SANS algorithm search process without workload concerns.

SA and NS may present significant solution improvements.

6.3. SANS algorithm with workload concerns

The ability of the SANS algorithm to comply with workload concerns (maximum shift duration and route workload balance) is analyzed by comparing the solutions obtained using this algorithm without and with these concerns. Instances with 50 or 100 bins require only one route that takes less time than the maximum shift duration (7 h), so these concerns are only relevant for the 250-bin and 500-bin instances. For these instances, if workload concerns are not considered, there are shift duration violations and duration imbalance between routes. This can be seen in Table 7 for the SANS algorithm results regarding 500-bin instances. For most of the routes, there are maximum shift duration violations, and great imbalances between the shortest and the longest routes, as they vary between 260.7 min (more than 4 h) and 477.0 min (almost 8 h).

Table 7

Average route duration indicators for 500-bin instances without workload concerns.

Instance	Number of routes	Route duration (minutes)			Shift duration violations	
		Longest	Shortest	Difference	Number of routes	Maximum violation (minutes)
1	6	579.0	305.0	274.0	5	159.0
2	6	643.8	166.2	477.0	5	223.8
3	6	634.9	264.8	370.1	4	214.9
4	6	613.3	352.6	260.7	5	193.3
5	6	578.5	307.3	271.2	5	158.5

Regarding the exact algorithm results, maximum shift duration violations and workload imbalance among collection teams are also a reality for all the 250-bin instances to which solutions were found. As an example, in instance 1, two of the three routes take 525 min and 550.7 min and the difference between the longest route and the shortest route is equal to 160.8 min (more than 2.5 h).

After integrating workload concerns in the SANS algorithm, the results obtained for the same indicators as well as the individual route durations are shown in Appendix F. As expected, the results show that the SANS algorithm is able to comply with the aforementioned concerns. The differences between the longest route and the shortest route is always lower than 15 min, which signifies an almost perfect route workload balance. There are still maximum shift duration violations in some cases, but these are very small (smaller than 5 min for every case). This can be easily solved by increasing the penalty related to this violation in the objective function. However, these small violations are acceptable

and the trade-off between maximum shift duration violations and profit is the most favorable one with this penalty equal to 0.5.

The results for the 250-bin instances (see [Appendix F](#)) are very similar to the ones obtained for the 500-bin instances. The SANS algorithm is able to ensure a good route workload balance, being the maximum difference between the longest route and the shortest route for each instance and run equal to 5.9 min. With respect to maximum shift duration violations, they exist in a greater number of routes than in the 500-bins instances, being smaller than 3.7 min for all cases except two in which they are equal to 5.2 and 5.9 min. The reasons for accepting this were already explained before.

This analysis allows concluding that the SANS algorithm is able to successfully integrate workload concerns. The addition of such concerns to the problem is expected to affect the profit associated to the waste collection. In [Table 8](#), the results of the SANS algorithm with and without workload concerns for the best and the worst runs and each instance of 250 and 500 bins are shown. Results for all the runs can be seen in [Appendix E](#). For 250-bin instances, [Table 8](#) shows the profit decreases when workload concerns are taken into consideration. Comparing the best solutions found (with and without workload concerns), profit decreases, on average, 32.6% while the runtime doubles (from 1 h to 2 h) when workload concerns are introduced. The differences between the best solutions and the worst solutions found are higher when workload concerns are included, which indicates that they introduce greater solution variability. With respect to the 500-bin instances, [Table 8](#) demonstrates that the profit decreases, on average, 16.0% when workload concerns are considered (for the best solutions found). When workload concerns are included, the runtime doubles. For 500-bin instances, the 2-hour runtime limit is already reached without the inclusion of these concerns. So, when they are included, the number of SA iterations as well as the number of NS procedures repetitions need to be reduced to still comply with the runtime limit. The reduction of the procedures' repetitions may also have an influence on the decrease in the profit obtained and not only the inclusion of the workload concerns.

In [Fig. 8](#), examples of routes without and with workload concerns for 250-bin are shown. The differences between incorporating or not these concerns are notorious and involve the number of routes needed and the size of these routes. To comply with these concerns, there are routes that become shorter and routes that become longer, as expected. Moreover, routes that clearly correspond to longer traveling distances contain less bins. As previously said, this happens, because workload concerns are more limiting than the capacity of the vehicles considering the

parameters used in this analysis.

With respect to the role played by SA and NS in the search process, it is shown in [Fig. 9](#) that NS performed significantly better than SA for both 250- and 500-bin instances when workload concerns were included. In this case, despite NS performing much better than SA, it was concluded that SA plays an important role in identifying intermediate solutions that allow NS to find better final solutions. [Fig. 9](#) also shows that, in some iterations, solutions are worse than in previous iterations. This happens because, in this figure, we are considering the profit without penalties, while accepting or rejecting a solution is based on the profit with penalties (i.e., the value of the objective function). Concluding, improvements in profit with penalties may mean deteriorations in the actual profit, which is expected since penalties translate the existence of additional concerns.

7. Real-world study

The case study presented in this paper is the same as in [Ramos et al. \(2018a\)](#). It concerns the waste collection operations carried out by Valorsul – Valorização e Tratamento de Resíduos Sólidos, S.A. This company operates in the region of Lisbon, the capital city of Portugal. Valorsul is responsible for the collection of different types of waste including recyclables (glass, paper/cardboard, and plastic/metal). The collection of recyclables is performed in 14 municipalities by a homogeneous vehicle fleet based on a single depot located in Vale Covo, Bombarral (north of Lisbon). The focus of our study was paper/cardboard collection.

This section is divided into two subsections. The first one describes the data we used while the second one presents the KPIs obtained after applying the hybrid metaheuristic (Phases 1 and 2) to this case study and a comparison between these KPI values and the KPIs actually obtained by Valorsul.

7.1. Study data

For each type of recyclables, Valorsul routes are pre-defined and performed periodically without taking into consideration bin fill levels. Information on these levels exist because each time a bin is collected, the collection team registers them according to five possible categories: empty (0%), less than half (25%), half (50%), more than half (75%) and full (100%). However, the accuracy of the registers is recognized to be low.

Table 8
SANS algorithm results with and without workload concerns.

Instance size	Workload constraints	Instance	Run	Profit (€)	Runtime (seconds)	Instance size	Workload constraints	Instance	Run	Profit (€)	Runtime (seconds)
250 bins	No	1	Best	344.53	3343	500 bins	No	1	Best	919.77	7200
			Worst	302.99	3548				Worst	879.51	7200
		2	Best	281.29	3540			2	Best	743.57	7200
			Worst	253.13	3226				Worst	702.25	7200
		3	Best	318.28	3570			3	Best	843.82	7200
			Worst	295.45	3254				Worst	811.68	7200
		4	Best	349.88	2893			4	Best	829.41	7200
			Worst	311.20	3077				Worst	781.28	7200
		5	Best	361.36	3120			5	Best	883.55	7200
			Worst	330.21	3148				Worst	848.70	7200
	Yes	1	Best	238.39	6426		Yes	1	Best	820.41	7200
			Worst	182.57	6625				Worst	799.07	7200
		2	Best	194.33	6636			2	Best	601.45	7200
			Worst	116.32	6478				Worst	550.44	7200
		3	Best	243.39	6213			3	Best	783.15	7200
			Worst	200.16	6302				Worst	733.47	7200
		4	Best	189.38	6560			4	Best	631.45	7200
			Worst	124.67	6700				Worst	606.91	7200
		5	Best	246.84	6404			5	Best	714.23	7200
			Worst	199.31	6468				Worst	674.09	7200
	Average (%)		Best	32.6	48.9		Average (%)		Best	16.0	0.0
			Worst	45.1	50.1				Worst	16.7	0.0

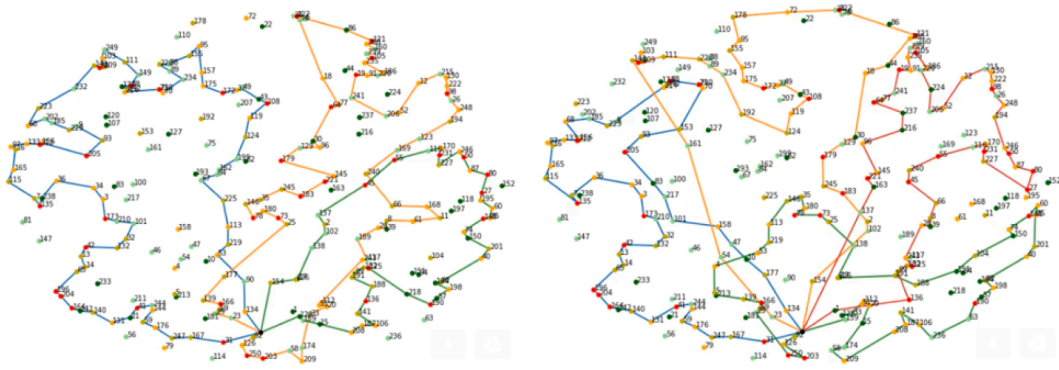


Fig. 8. Solutions of 250-bin instance 4 without (left) and with workload concerns (right).

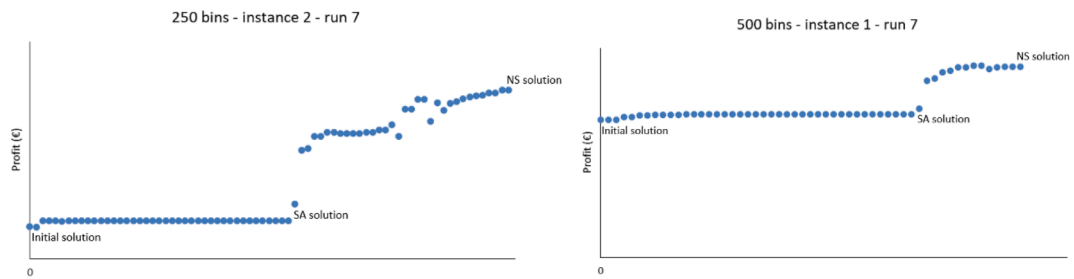


Fig. 9. SANS algorithm search process with workload concerns.

The collection of paper/cardboard (our focus) is carried out in 26 periodic pre-defined routes, of which only 3 are considered here (as in Ramos et al., 2018a): route 6 with 68 bins, route 11 with 74 bins and route 13 with 84 bins, for a total of 226 bins. Every time a route is performed, all the bins in the route are visited even if they may be empty or with a low fill level. A planning period of 30 days (in January) was considered in our case study. The location of the depot (black) and of the bins (blue) included in routes 6, 11, and 13 is presented in Fig. 10.

The scalar data for the study are the same as the ones shown in Table 4 (Subsection 5.1), except for B that is equal to 226 bins. Other data used were the (shortest) travel distances between the depot and the bins and between the bins (d_{ij}), and the expected accumulation rate per bin i at day t (a_{it}). This accumulation rate was calculated by dividing the bin fill levels registered by the collection team by the number of days that passed after the previous day the bins were collected.

The key performance indicators (KPIs) corresponding to Valorsul

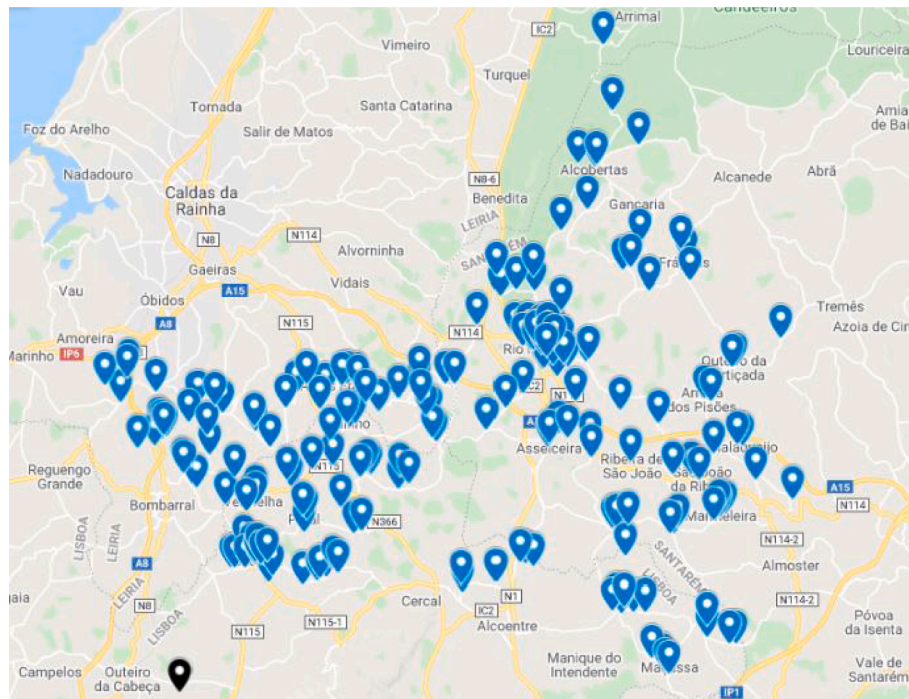


Fig. 10. Location of depot and bins of routes 6, 11 and 13.

operations for routes 6, 11 and 13 in the period analyzed are presented in Table 9. According to this table, the profit obtained by Valorsul from routes 6, 11, and 13 in the 30-day period under analysis was equal to 455€. A total of 8 routes was completed in that period (routes 6 and 11 were repeated twice and route 13 four times), and 620 bins were visited, of which around 12% were empty and 1% were overflowing. The waste collected from those bins amounted to 17,000 kg, and 1200 km were traveled, resulting in an average of around 15 kg per km traveled). The vehicle usage rate (total weight of waste in the vehicle divided by its capacity) varied between 46% and 67%. The maximum shift duration of 7 h was violated in 2 (out of 8) routes, in one case by almost 2 h (112 min) and in the other by less than 20 min. With respect to route workload balance, a significant imbalance of 111 min was observed in day 22, in line with the large violation of the maximum shift duration occurred in that day.

7.2. Study results

In this subsection, we present the KPIs that summarize the solutions obtained through the hybrid metaheuristic considering workload concerns to our case study. Due to the random nature of the metaheuristic, it has run 10 times. Given the characteristics of the case study, the combination of parameters used was the same as for the 250-bin test instances. It is important to note that, in this case, runtime does not represent a limitation, and therefore, 80,000 iterations were performed for each run. In Table 10, the KPIs for the best, the closest to the average and the worst runs of the metaheuristic are displayed (KPIs for the remaining runs are presented in Appendix G). In the next paragraphs, these KPIs are compared among them and with the KPIs of the actual Valorsul operations (Table 9).

It can be seen in Table 10 that even in the worst performing run (5), the metaheuristic leads to a profit 45% higher than that corresponding to the routes implemented by Valorsul (658.7€ against 455€). The best performing run (4) provides a profit of 727€, i.e., a profit gain of 60%. Profits thus vary between 659€ and 727€ (68€ difference) across all runs. The run whose profit (696€) is closer to the average profit for all runs (693€) is run 3.

Regarding traveled distances, the maximum distance was obtained for run 5 (the worst) and is equal to 1007 km, and the minimum distance, 956 km, was obtained for run 8. The maximum distance variation across runs is just around 60 km.

The number of bins visited in the metaheuristic runs is considerably lower than the number of bins actually visited by Valorsul (482 against 620). Empty bins are not visited in any run of the hybrid metaheuristic, while Valorsul visited 73 empty bins. The maximum number of overflowing bins obtained with the metaheuristic is 4, which is the same number as in Valorsul actual operations. However, with our metaheuristic, this only happened in four runs. In five runs, the number of overflowing was just 2 in the 30-day period.

Table 9
KPIs for the actual Valorsul operations.

Day	1	7	8		15	19	22	Total
Route	13	11	6 and 13		13	11	6 and 13	
KPI								
Profit (€)	118	94	39		126	75	5	455
Weight (kg)	2472	2343	1420	2564	2693	2251	1513	17,450
Distance (km)	117	129	208	132	130	139	221	1203
Visited bins	84	74	152		84	74	152	620
Overflowing bins	0	0	1		1	0	2	4
Empty visited bins	6	17	33		7	2	8	73
% of empty bins	7.1	23.0	21.7		8.3	2.7	5.3	11.8
Ratio (kg/km)	21.1	18.2	11.7		20.7	16.2	10.7	14.5
Vehicles used	1	1	2		1	1	2	8
Vehicle usage rate (%)	61.8	58.6	49.8		67.3	56.3	46.3	–
Shift duration (minutes)	410	364	438	428	427	424	532	3445
Longest shift – shortest shift (minutes)	–	–	10		–	–	111	–

The minimum kg/km ratio (17) achieved by the metaheuristic is also higher than the one associated to Valorsul operations (15). The number of routes needed during the period according to the results of the metaheuristic is 7 for every run. This represents an improvement regarding the Valorsul solution, in which 8 routes were needed.

Vehicle usage rates are always higher than 50%, varying between 51% (day 22 – run 2) and 86% (day 8 – run 3), when the hybrid metaheuristic is applied. Valorsul vehicle usage rates are lower, starting at 46% (day 22) and reaching 67% (day 15) at maximum.

Maximum shift duration constraints were either respected by the hybrid metaheuristic or violated by less than 5 min, which is acceptable according to the reasons previously explained in Section 5. There is only one exception to this: day 8 of run 3, for which a route of 435 min is performed. Nevertheless, this is much better than what Valorsul did, as one of its routes exceeded the maximum shift duration by almost 2 h.

Lastly, with respect to route workload balance objective, the hybrid metaheuristic was able to generate very balanced routes when more than one route was necessary, being the highest difference for every pair of routes in every run equal to only 2 min. This is also better when compared to the Valorsul's actual routes.

Runtimes varied between 1274 s (around 20 min) when only one route was needed and 2994 s (approximately 50 min) when two routes were needed. These runtimes are perfectly viable in practice.

8. Conclusion

This paper presents a new hybrid metaheuristic to solve large-size instances of the smart waste collection routing problem with workload concerns. Several decisions are at stake: in which days collection is necessary; if collection is necessary, which are the must-go bins (i.e., the bins to visit, because they are full, overflowing, or forecasted to overflow before the following collection day); and the best route(s) to visit these bins as well as other bins that may improve the route(s) profit if visited. The metaheuristic takes into consideration historical data regarding bins' accumulation rates, and looks into the (expected) future to better plan collection. The bins that may create more profitable routes if visited and the best collection routes to perform are obtained through a simulated annealing/neighborhood search (SANS) algorithm. Development, calibration and an extensive computational study of this metaheuristic considering different size instances (50, 100, 250 and 500 bins) are presented in this paper.

The results of our study show that, unlike exact optimization algorithms, our hybrid metaheuristic is able to handle large-size instances, leading to good-quality solutions that comply with the needs of practice (including maximum shift duration and route workload balance) within acceptable runtimes (two hours). For small size instances, with 50 and 100 bins, the hybrid metaheuristic reached profits that are, on average, 0.7% and 3.5%, respectively, far from the ones produced by the exact method. When applied to a real case study involving recyclables

Table 10
KPIs for the solutions obtained through the hybrid metaheuristic.

Run	4 – The best								
Day	1	8	15	22	Total	Δ			
Profit (€)	217	156	227	126	727	272			
Weight (kg)	3060	1930	3204	3107	2070	2653	1700	17,724	274
Distance (km)	106	151	148	113	152	135	152	956	–247
Visited bins	133	67	144	137	481	–139			
Overflowing bins	0	2	0	0	2	–2			
Empty visited bins	0	0	0	0	0	–73			
Ratio (kg/km)	19	22	20	15	19	4			
Vehicles used	2	1	2	2	7	–1			
Vehicle usage rate (%)	62	80	65	54	–	–			
Shift duration (minutes)	392	391	423	415	414	421	421	2877	–568
Longest shift – shortest shift (minutes)	1	–	1	1	–	–			
Runtime (seconds)	2310	1369	2383	2099	–	46672513335000–			
Run	3 – The closest to average								
Day	1	8	15	22	Total	Δ			
Profit (€)	186	181	206	123	696	240			
Weight (kg)	3207	1772	3421	2743	2060	2831	1603	17,638	188
Distance (km)	119	169	144	112	138	126	172	979	–223
Visited bins	131	73	139	131	474	–146			
Overflowing bins	0	3	0	0	3	–1			
Empty visited bins	0	0	0	0	0	–73			
Ratio (kg/km)	17	24	19	15	18	4			
Vehicles used	2	1	2	2	7	–1			
Vehicle usage rate (%)	62	86	60	55	–	–			
Shift duration (minutes)	412	412	435	397	396	420	420	2891	–554
Longest shift – shortest shift (minutes)	1	–	1	0	–	–			
Runtime (seconds)	2023	1346	2307	2157	–	46672514287500–			
Run	5 – The worst								
Day	1	8	15	22	Total	Δ			
Profit (€)	200	168	202	89	659	203			
Weight (kg)	3192	1918	3264	3006	1971	2474	1715	17,540	89
Distance (km)	121	163	142	120	151	146	162	1007	–196
Visited bins	139	69	144	126	478	–142			
Overflowing bins	0	4	0	0	4	0			
Empty visited bins	0	0	0	0	0	–73			
Ratio (kg/km)	18	23	18	14	17	3			
Vehicles used	2	1	2	2	7	–1			
Vehicle usage rate (%)	64	82	62	52	–	–			
Shift duration (minutes)	422	422	421	420	419	420	421	2944	–501
Longest shift – shortest shift (minutes)	0	–	1	0	–	–			
Runtime (seconds)	2099	1328	2076	1929	–	–			
Δ = Total(run) – Total(Valorsul)									

collection by Valorsul (the waste collection company of the Lisbon region), even the worst run of the metaheuristic led to profits 45 % higher than the ones currently obtained by Valorsul (it is important to note that the metaheuristic has a random nature, and therefore solutions often change from run to run). Improvements were also found in all other KPIs considered to assess Valorsul collection operations as well as in the compliance with workload concerns (maximum shift duration and route workload balance).

Despite being useful as currently implemented both for waste collection companies and for other companies dealing with smart collection problems, the proposed hybrid metaheuristic can be improved with respect to a number of issues. We focus here on two. The first has to do with parameter calibration. Notwithstanding the effort we put into this task, we recognize that applying an automated design approach can lead to a better choice of parameter values (Stützle and López-Ibáñez, 2019). The other issue is related with uncertainty. We have worked with average waste accumulation rates and expected profits, weighted by a safety margin, and therefore their variability was considered in a straightforward way. This variability consideration may be improved in several different ways, depending, e.g., on how scenarios are generated (and, possibly, reduced) and on how it is dealt with in the objective function (and/or in the constraints). These are important issues that we will certainly address in our future research in waste collection

problems.

CRedit authorship contribution statement

Diana Jorge: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing - original draft, Writing - review & editing, Visualization. **António Pais Antunes:** Conceptualization, Methodology, Validation, Writing - review & editing, Supervision. **Tânia Rodrigues Pereira Ramos:** Conceptualization, Methodology, Validation, Writing - review & editing, Supervision. **Ana Paula Barbosa-Póvoa:** Conceptualization, Methodology, Validation, Writing - review & editing, Supervision.

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the needs and possibilities of practice.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cor.2021.105518>.

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